

Pablo García Martínez

Spain (Open to relocation across EMEA) | pablogarmar04@gmail.com | github.com/garmaar | LinkedIn

Availability: Summer 2027 (June – September, 13+ weeks)

PROFILE

Telecommunications Engineering student focused on Site Reliability Engineering, distributed systems and cloud infrastructure. Experienced building AWS serverless architectures, automation pipelines and network performance models using Python, Terraform and machine learning. Focused on designing scalable, reliable and secure systems.

EDUCATION

Universidad de Sevilla (ETSI)

Bachelor's Degree in Telecommunications Engineering

Politecnico di Milano

Erasmus+ Academic Exchange – Telecommunications Engineering (Networking Focus)

UCAM Murcia

SICUE Mobility Program (Telecommunications Engineering)

Seville, Spain

Expected July 2028

Milan, Italy

Completed

Murcia, Spain

Enrolled

TECHNICAL SKILLS

Programming: Python, C, Bash, Java, FastAPI

Cloud & Infrastructure: AWS, Terraform, Docker, Infrastructure as Code (IaC)

AWS Services: S3, CloudFront, Lambda, API Gateway, DynamoDB, IAM, STS

Systems & Networking: Linux, TCP/IP, HTTP, DNS, Routing, Subnetting, OSPF, BGP, OSI Model

Data & Machine Learning: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib

CI/CD & Tools: GitHub Actions, Git, Linux CLI

SYSTEMS & ENGINEERING COURSEWORK

Operating Systems, Computer Networks, Distributed Systems, Probability and Statistics, Queueing Theory, Network Performance Analysis, Machine Learning.

TECHNICAL PROJECTS

Serverless AWS Resume Platform (Cloud Resume Challenge) | AWS, Terraform, GitHub Actions, Python

- Built a serverless AWS application following production cloud architecture practices using Amazon S3, CloudFront, Lambda, API Gateway and DynamoDB, provisioning infrastructure through Terraform.
- Automated infrastructure and application deployments using GitHub Actions with AWS OIDC federation, eliminating long-lived credentials and enabling secure CI/CD workflows.
- Developed a Python Lambda backend for a visit counter API and implemented IAM policies, CloudFront Origin Access Control and CloudWatch monitoring for secure and reliable operation.

5G Throughput Prediction & Federated Learning | Python, PyTorch, Scikit-learn

- Developed a machine learning pipeline to predict 5G network throughput using a dense spatial dataset with 100k+ measurements including SINR, PRB allocation, BLER, traffic type and user features.
- Implemented a Federated Averaging (FedAvg) training pipeline in PyTorch simulating decentralized 5G user nodes, comparing convergence and prediction performance against centralized baselines.
- Applied time-aware validation strategies, including temporal train/test splitting and TimeSeriesSplit, preventing data leakage during network performance prediction.

Polling Systems: Discrete-Event Simulation | Python, SimPy, NumPy, Pandas

- Developed a modular discrete-event simulator to evaluate polling-based multiple access systems under Poisson traffic loads and different service disciplines.
- Validated simulation outputs against analytical queueing models with relative errors below 1%, analyzing throughput, latency and fairness trade-offs under asymmetric network congestion.

CERTIFICATIONS & ACHIEVEMENTS

Awards: Ranked 63rd Overall in the Google Cloud EMEA University Challenge

Certifications: AWS Certified Solutions Architect – Associate (Expected December 2026)

Languages: English (C1 Certified, Universidad de Sevilla), Italian (Intermediate), Spanish (Native)